Q451 (Task 0): the winding sectors of the cylindric Jacobi–Trudi formula

Each sector is ± a *single* Schur function — so the pre-registered trigger fires and the inference attached to it does not follow

Clio

2026-10-14 **(PROVE c1)** **For:** Robin Langer **cc:** Rick
Repository: `clio-claude/proofs`; **base commit** `b82a04f`

The question, and why it was asked

HKKO `2301.13117`, Thm **??** `thm:ssyt2` (eq. `eq:CJT`, line 877 of the source, *published and uncommented*) states, for $\lambda \in \mathrm{Par}(h, w)$,

$$s_{\lambda[h,w]'}(\mathbf{x}) = \sum_{\substack{k_1,\dots,k_h \in \mathbb{Z} \\ k_1+\cdots+k_h=0}} \det_{1\le i,j\le h}\Big(e_{\lambda_i - i + j + (h+w)k_i}(\mathbf{x})\Big).$$

The 10-14 WAKE finding established that HKKO's six `%`-commented "κ-graded" blocks are this identity substituted into their own live statement — a macro, not a suppressed theorem. Q451 then asks the follow-on: is the κ-grading destroyed later, at Gordon's Lemma 4.4, or was it **never well defined**, because κ is merely the summation index of a determinant formula for an object that is already the sum over κ?

What was measured

Each sector $\det_h\big(e_{\lambda_i - i + j + (h+w)k_i}\big)$ was computed in Λ_n (pure Python + SymPy; SageMath is not installed) and expanded in the Schur basis via dual Jacobi–Trudi plus an exact linear solve.

Positive control (two mechanisms, no shared code path). The full sector sum was compared against a **direct lattice-path enumeration** — HKKO `prop:cssyt=path`, $K^{\alpha}_{\lambda[h,w]'} = \#\{\text{paths } 0 \to \lambda \text{ in } \{x_1 \ge \cdots \ge x_h \ge x_1 - w\}\}$ — giving $s_{\lambda[h,w]'} = \sum_\alpha K^\alpha \mathbf{x}^\alpha$. 10 **of** 10 **exact**, over $h \in \{2,3\}$, $w \in \{2,4\}$, |CRST| printed in each case (from 0 to 74 tableaux).

Negative control. A deliberately wrong modulus $h+w \to h+w+1$ must break the reconstruction. It fired on 2 **of** 4 fibres — precisely the two where a $k \neq 0$ sector is active at the correct modulus. On the other two only $k = 0$ contributes at either modulus, so the control is *structurally blind* there. That is reported rather than averaged away.

Proposition 1 (measured, $h \in \{2,3\}$, $w \in \{2,4\}$, 10 shapes, 11 nonzero sectors)**.** *Every nonzero sector is ± a* ***single*** *Schur function. The $k = 0$ sector is $+s_{\lambda'}$ (ordinary, by dual Jacobi–Trudi); every sector with $k \neq 0$ carries a* ***negative*** *coefficient. Example: $h = 2$, $w = 2$, $\lambda = (3,1)$ gives $s_{\lambda[2,2]'} = s_{(2,1,1)} - s_{(1,1,1,1)}$, with* $|\mathrm{CRST}| = 14$.

The verdict, unbundled

Pre-registered outcome **(0A)** triggers on "the k-terms are not individually Schur-positive, or degenerate". The first disjunct **fired**: sectors with $k \neq 0$ are Schur-*negative*. But (0A)'s attached inference — "the grading has nothing to grade, so it dies at Jacobi–Trudi, before Gordon; close Q448" — **does not follow** from what was measured, for a reason the pre-registration did not anticipate: each sector is $\pm$ one Schur function, and distinct sectors landed on distinct partitions, so on this range the decomposition is *recoverable from the object itself* (Schur functions are linearly independent). A decomposition recoverable from the object is not a formula artifact.

A pre-registered trigger and the inference bolted to it can come apart. Report the trigger; re-der

Therefore: Q451 is answered *negatively* — the grading does not die at the Jacobi–Trudi expansion. **Q448 is not closed by this.** What the sectors are not is *positive*: they are signed reflection corrections, so a t-grading indexed by κ grades a signed decomposition, not a tableau statistic. We do **not** claim Gordon's involution is the obstruction: Gordon was not tested here. Measured range: $h \leq 3$, $w \leq 4$, $|\lambda| \leq 7$, $n \leq 4$ variables; a sector collision (two k on one μ) would need a larger n than was reached, and was not observed.